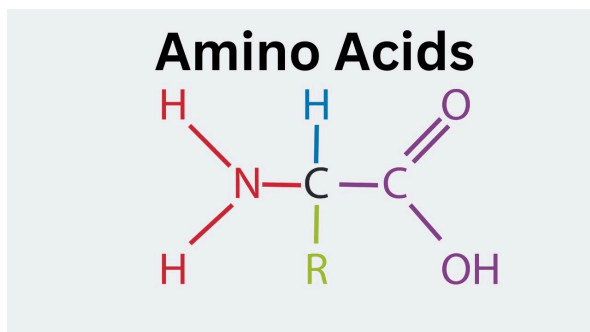


Proteins Study Guide

B1.2.1— Generalized structure of an amino acid. *Students should be able to draw a diagram of a generalized amino acid showing the alpha carbon atom with amine group, carboxyl group, R-group and hydrogen attached.*

- Draw the generalized structure of an amino acid.



- Label the amine group, carboxyl group, alpha carbon and R group on an amino acid.

Amine- has NH_2

Carboxyl- has $-\text{COOH}$

R- anything else, variable side chain attached to middle C (alpha carbon)

B1.2.2— Condensation reactions forming dipeptides and longer chains of amino acids.

Students should be able to write the word equation for this reaction and draw a generalized dipeptide after modeling the reaction with molecular models.

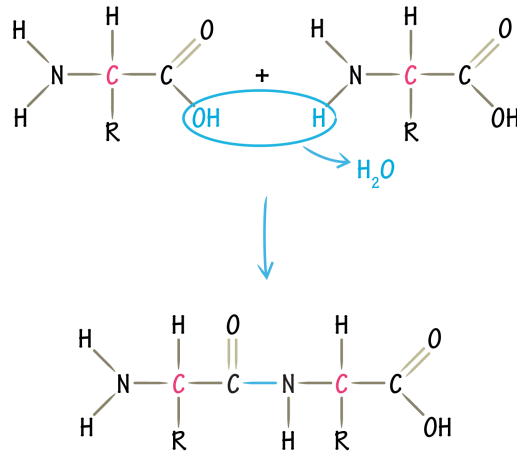
- Define dipeptide and polypeptide.

Dipeptide: 2 amino acids together by a single peptide bond

Polypeptide: more than two amino acids together by peptide bond

- Draw peptide bond formation in a condensation reaction between two amino acids.

Peptide Bond Formation



- State where in the cell polypeptide formation occurs.

Ribosome

B1.2.3-- Dietary requirements for amino acids. *Essential amino acids cannot be synthesized and must be obtained from food. Non-essential amino acids can be made from other amino acids. Students are not required to give examples of essential and non-essential amino acids. Vegan diets require attention to ensure essential amino acids are consumed.*

- Compare the source of amino acids by plant and animal cells.

Plants: can synthesize all amino acids

Animals: need to eat in order to get all amino acids

- Define “essential” and “non-essential” as related to dietary amino acids.

Essential = we need to eat

Nonessential = produce ourselves

- Outline why vegan diets require attention to food combinations to ensure essential amino acids are consumed.

Since they don't get the nutrients they need that come from animals because plants don't have all 9 essential amino acids that humans need

B1.2.4— Infinite variety of possible peptide chains. *Include the ideas that 20 amino acids are coded for in the genetic code, that peptide chains can have any number of amino acids, from a few to thousands, and that amino acids can be in any order. Students should be familiar with examples of polypeptides.*

- Outline why there is a limitless diversity of DNA base sequences.

20 different amino acids, polypeptides can contain any number of amino acids

Amino acids can be arranged in any sequence therefore, an enormous number of different proteins can be produced

B1.2.5— Effect of pH and temperature on protein structure. *Include the term “denaturation”*

- Define denaturation.

Disrupting secondary, tertiary, or quaternary structure (unfolding of structure)

- Explain the effect of pH on temperature on protein structure and function.

Since temperature (heat) makes the proteins vibrate, intermolecular bond is broken

Since pH is either acidic or alkaline, charged R groups get their ionic bonds broken in protein

B1.2.6- Chemical diversity in the R-groups of amino acids as a basis for the immense diversity in protein form and function. *Students are not required to give specific examples of R-groups. However, students should understand that R-groups determine the properties of assembled polypeptides. Students should appreciate that R- groups are hydrophobic or hydrophilic and that hydrophilic R-groups are polar or charged, acidic or basic.*

- Outline the effect of R-group structure on the properties of an amino acid, with reference to hydrophilic, hydrophobic, polar and charged.

If R group has no O or S, = nonpolar = hydrophobic

If R group has O or S, = polar = hydrophilic

If R group is charged (+/-) = charged

B1.2.7— Impact of primary structure on the conformation of proteins. *Students should understand that the sequence of amino acids and the precise position of each amino acid within a structure determines the three-dimensional shape of proteins. Proteins therefore have precise, predictable and repeatable structures, despite their complexity.*

- Identify the “backbone” of a polypeptide.

-N-C-C

- Define “conformation” as related to protein structure.

The 3d arrangement of atoms in a polypeptide/protein
determines function of proteins

- Describe the primary structure of a protein, including the type of bonding involved.

Primary: sequence chain

Bond: peptide bond

- Outline how a DNA sequence codes for a polypeptide that will repeatedly fold into the same precise, predictable protein confirmation.

DNA sequence codes the same sequence of amino acids, they will fold into same way bc of peptide bonding

DNA = transcription , translation, → amino acid sequence → folded into amino acid

B1.2.8- Pleating and coiling of secondary structure of proteins. *Include hydrogen bonding in regular positions to stabilize alpha helices and beta-pleated sheets.*

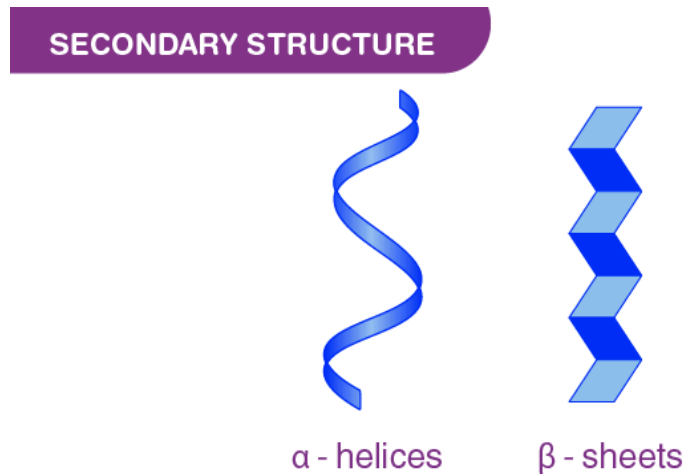
- Describe the secondary structure of a protein, including the type and location of the bonds involved.

Secondary: beta pleated sheet, alpha helix

Bond: hydrogen bond

Location of when bond occurs: polypeptide backbone -N-C-C

- Identify the alpha-helix and beta-pleated sheet in images of protein structure.



B1.2.9- Dependence of tertiary structure on hydrogen bonds, ionic bonds, disulfide covalent bonds and hydrophobic interactions. *Students are not required to name examples of amino acids that participate in these types of bonding, apart from pairs of cysteines forming disulfide bonds. Students should understand that amine and carboxyl groups in R-groups can become positively or negatively charged by binding or dissociation of hydrogen ions and that they can then participate in ionic bonding.*

- Describe the tertiary structure of a protein, including the types of R-group interactions involved.

Tertiary: 3D shape of protein

Bond: ionic between + and - charged R groups, hydrogen bonds, disulfide bridges, hydrophobic interactions

- Explain the effect of polar and non-polar R-groups of amino acids on tertiary structure of proteins.

Nonpolar inside the tertiary protein structure

Hydrophobic R-groups inside while Hydrophilic R groups outwards

- Explain the effect of positively and negatively charged amino acid R-groups on the tertiary structure of proteins.

Ionic bondings

Determines how a protein is folded and determines biological function

- State that a strong disulfide covalent bond can occur between pairs of cysteine amino acids.

B1.2.10- Effect of polar and non-polar amino acids on tertiary structure of proteins.

In proteins that are soluble in water, hydrophobic amino acids are clustered in the core of globular proteins.

Integral proteins have regions with hydrophobic amino acids, helping them to embed in membranes.

- State that amino acids can be hydrophobic or hydrophilic depending on the properties of the R-group
- Discuss the arrangement of amino acids in soluble globular proteins.

Hydrophilic on the outside, hydrophobic on the inside

- Discuss the arrangement of amino acids in integral membrane bound proteins.

Hydrophobic facing lipid bilayer in membrane

Hydrophilic located on extra cellular and cytoplasmic surfaces

- Discuss the arrangement of amino acids in channel proteins in membranes.

Hydrophobic facing lipid bilayer

Hydrophilic on line the interior of the channel pore

B1.2.11- Quaternary structure of non-conjugated and conjugated proteins.

Include insulin and collagen as examples of non-conjugated proteins and hemoglobin as an example of a conjugated protein.

NOS: Technology allows imaging of structures that would be impossible to observe with the unaided senses. For example, cryogenic electron microscopy has allowed imaging of single-protein molecules and their interactions with other molecules.

- Describe the quaternary structure of a protein.

Multiple polypeptides combine

- Compare the structure of conjugated and non-conjugated proteins.

Nonconjugated only amino acid chains

Conjugated has additional non-protein

- State an example of a conjugated and non-conjugated protein.

Conjugated: hemoglobin

Nonconjugated: collagen

B1.2.12- Relationship of form and function in globular and fibrous proteins. *Students should know the difference in shape between globular and fibrous proteins and understand that their shapes make them suitable for specific functions. Use insulin and collagen to exemplify how form and function are related.*

- Describe, with reference to collagen, the structure and function of fibrous proteins.

Long and elongated molecule like collagen has 3 polypeptides intertwined together

Provide strength, support, stability to cells + tissues

- Describe, with reference to insulin, the structure and specificity of globular proteins.

Compact, spherical protein folded structure formed by multiple polypeptide

Insulin specific to insulin receptor, allows specific and unambiguous signals to be sent to body cells when blood sugar concentration is too high

D1.2.18- Modification of polypeptides into their functional state. *Students should appreciate that many polypeptides must be modified before they can function. The examples chosen should include the two-stage modification of pre-proinsulin to insulin.*

- List types of modifications of polypeptides that may be required to form a functional protein.

Chemically altering R groups

Folding the protein

Combining proteins with other proteins or non proteins molecules

- Outline the two stages of modification of preproinsulin to form functional insulin.
 1. Signal peptide tells cell where to send insulin and get removed at the destination (proinsulin)
 2. 3 disulfide bridges form to stabilize tertiary structure
 3. Middle of protein remained now is at final form (insulin)

D1.2.19- Recycling of amino acids by proteasomes. Limit to the understanding that sustaining a functional proteome requires constant protein breakdown and synthesis. *Limit to the understanding that sustaining a functional proteome requires constant protein breakdown and synthesis.*

- List reasons when proteins typically exist for a relatively short time within a cell.

Damage, incorrect folding, regulatory control, changes in cell needs

They are recycled so body can repeat the process of translation and transcription using the old amino acid

- Outline the function of proteasomes in the recycling of amino acids.

Used at primary cellular machinery for recycling amino acids by breaking down unwanted or damaged protein into smaller peptides, further processed and reused to build new proteins